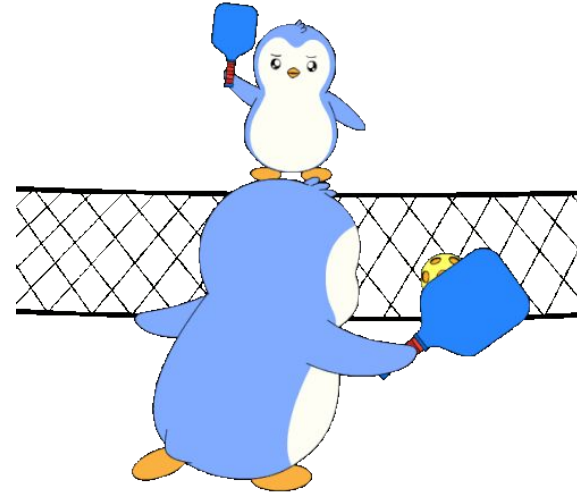




WTA Shots EDA Project

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WTA 2021- 2026 Data Overview



Dataset

- WTA match records from 2021-2026 (through May 25, 2026)
- 14,000+ matches
- ~50 original variables
- Each row represents one completed WTA match



Data Preparation

- Standardized inconsistent surface labels
- Converted winner/loser records into long format
- Filtered matches with missing data



Key Variables

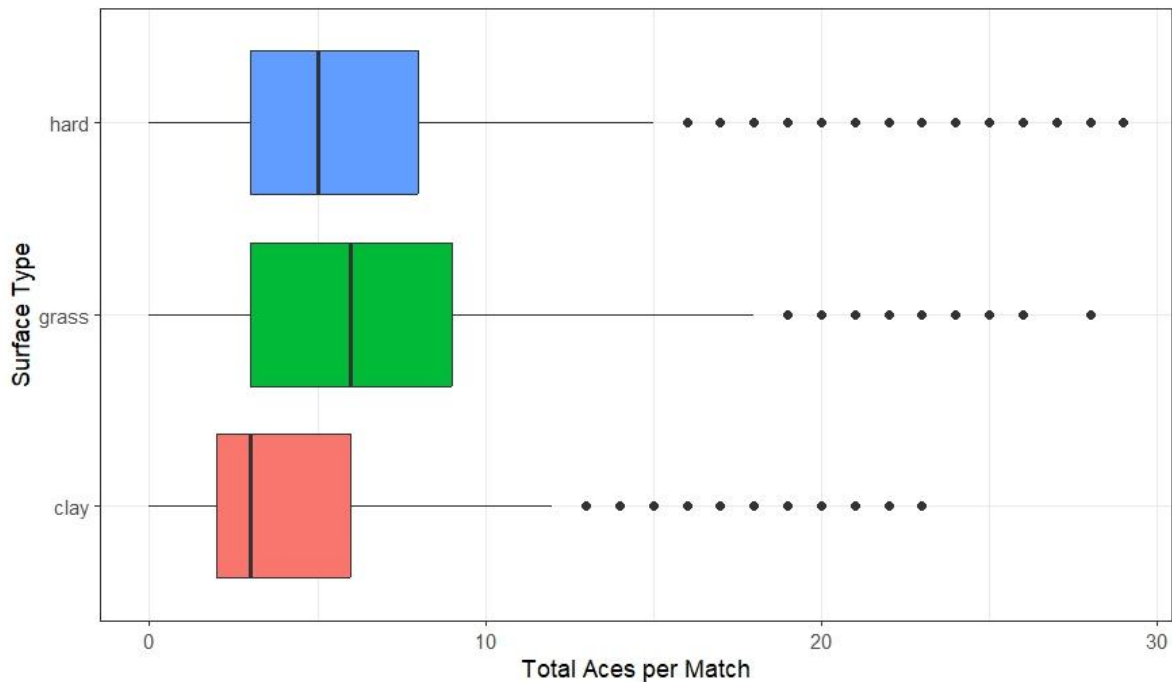
- **Player characteristics:** Ranking, Age, Height
- **Match information:** Tournament, Surface, Player Names
- **Performance statistics:** Aces, Double Faults, Serve Points Won

Question 1

Is the number of aces in a match affected by surface type and are these numbers affected by the distribution of surfaces by tournament level?



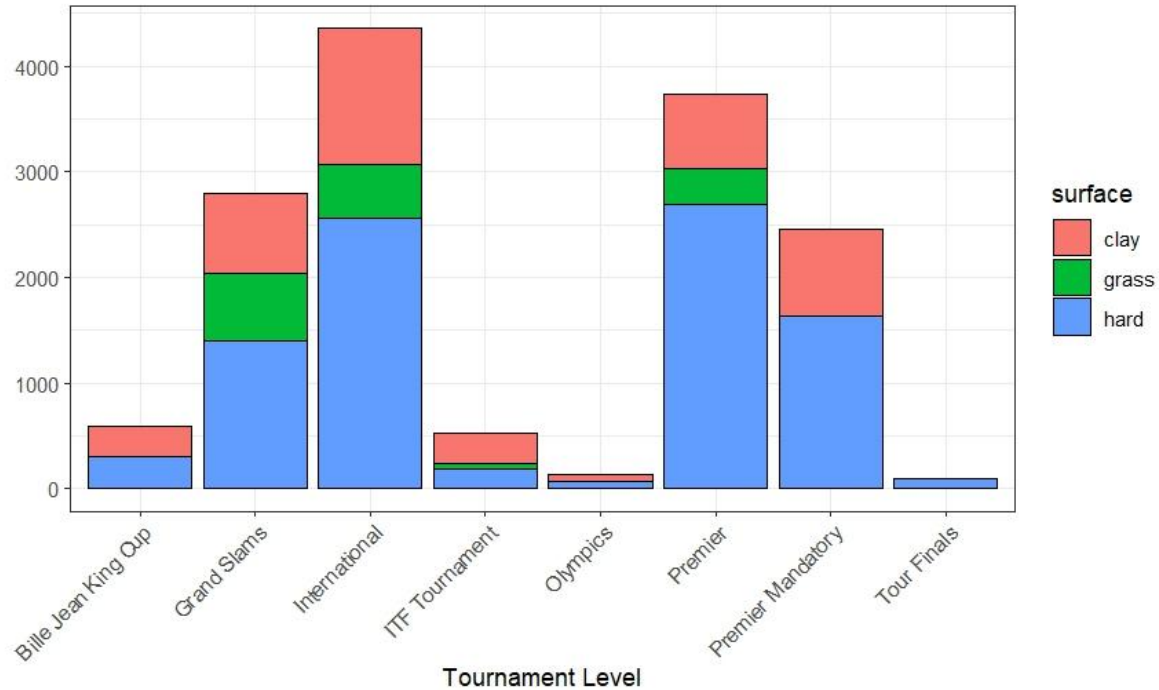
Distribution of Aces by Surface Types



Is there any significant difference between the surfaces?

Do these averages just have to do with the how frequently courts are played on?

Distribution of Surfaces by Tournament Level

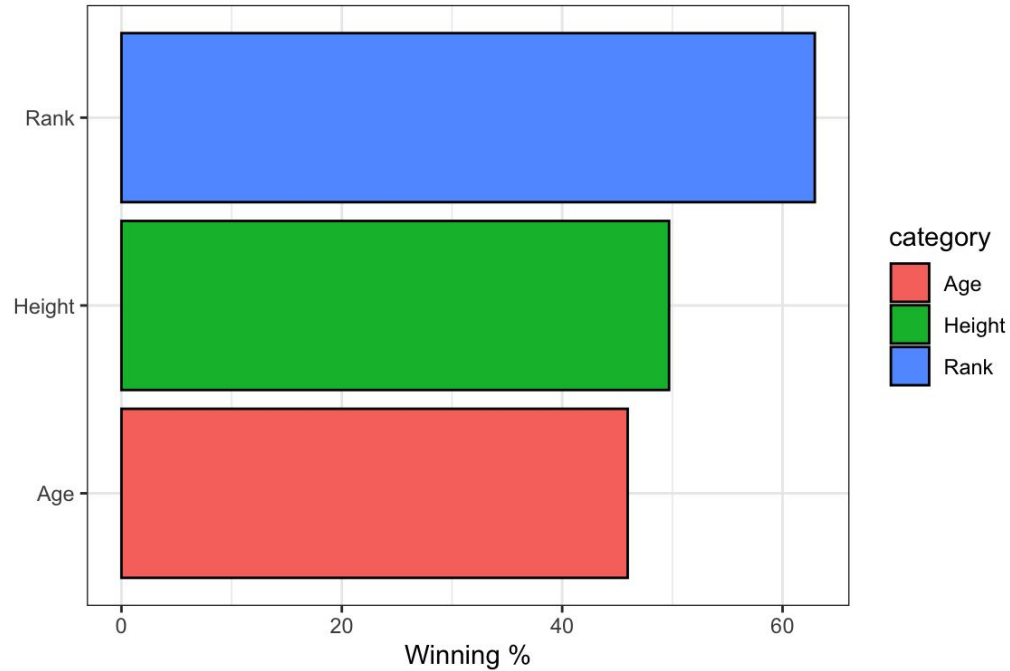


Question 2

Do Winning Players Usually Have a Ranking, Height, or Age Advantage?



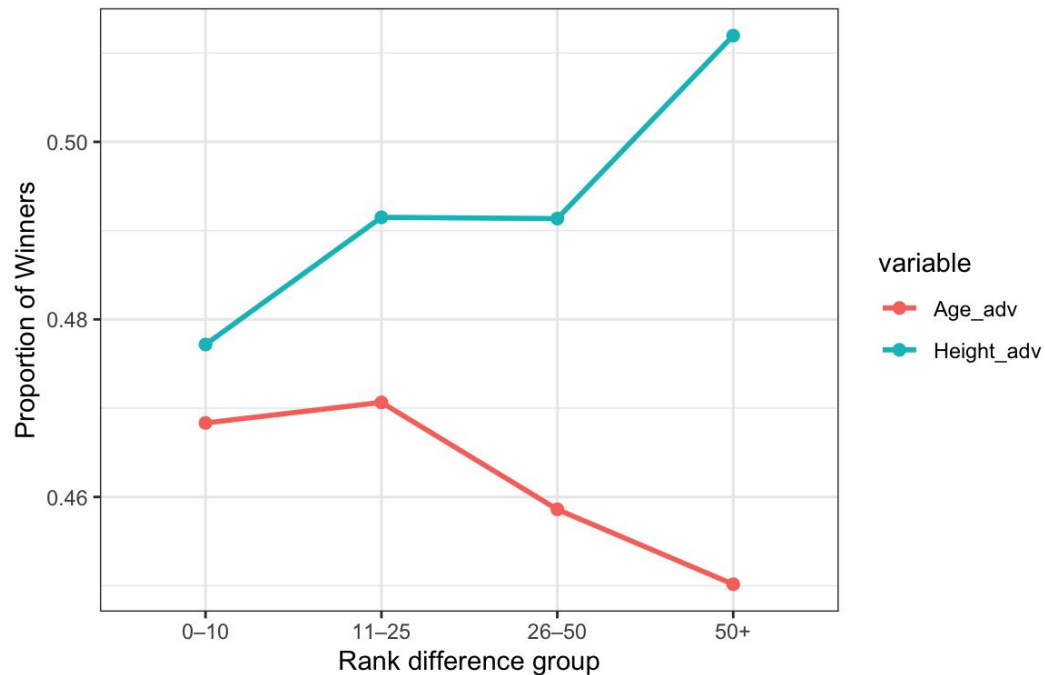
Winners had the better ranking in 63% of matches



Do Height and Age Matter More in competitive Matchups?



Height and Age Advantages in Close Matches



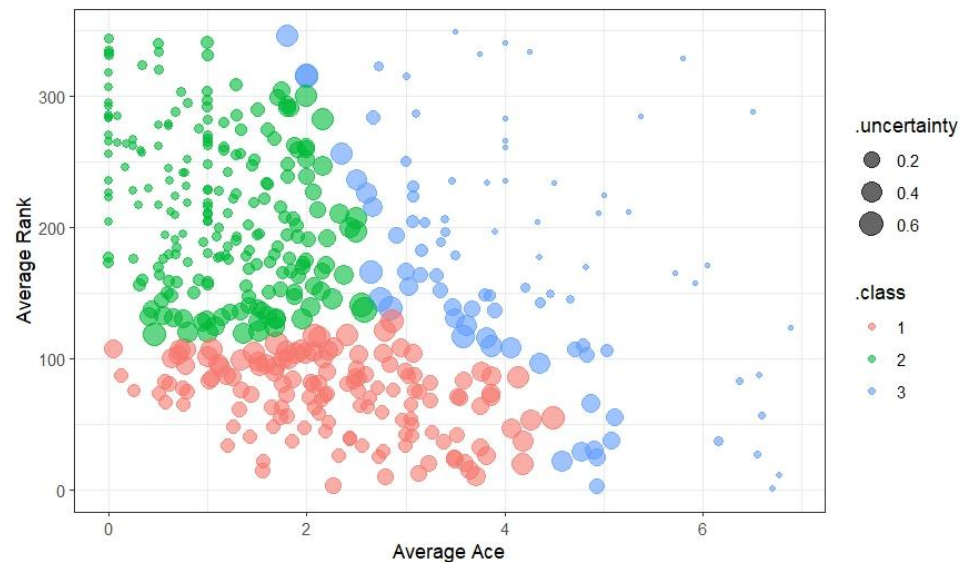
Height advantage becomes more common in less competitive matchups

Soft Clustering of Players using GMM

Clusters	Size	Proportion	Mean avg aces	Mean avg rank
1	142	0.2820672	2.311218	76.49255
2	215	0.4767004	1.156676	207.38534
3	90	0.2412324	3.706513	169.46996

Interpretation:

- ❑ **Elite Tier:** Highest-performing players
- ❑ **Low-Ace Majority:** Low serve impact, lower ranking
- ❑ **Serving Specialists:** High-ace servers (3.71 aces/match)



Summary & Takeaway

Question 1

The amount of aces per match are impacted by surface type not just the frequency a surface is played on.

Possible next steps:

Factor in players and potentially do a mixed model or repeated measures ANOVA.

Question 2

Higher-ranked players win most matches, regardless of height or age gaps.

Possible next steps:

Study combined effects, player progression over time and verify our observation with a statistical analysis.

Clustering

Fixed number of clusters assumed may not fit real player distributions.

Limited Scope: Using only two metrics omits rich point-level data.

Future Work: Integrate more performance variables and match context.

Questions?

